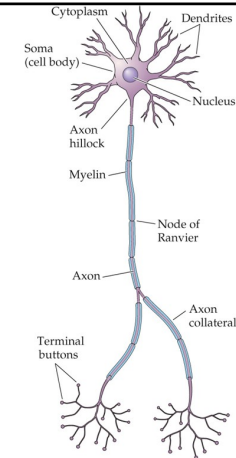


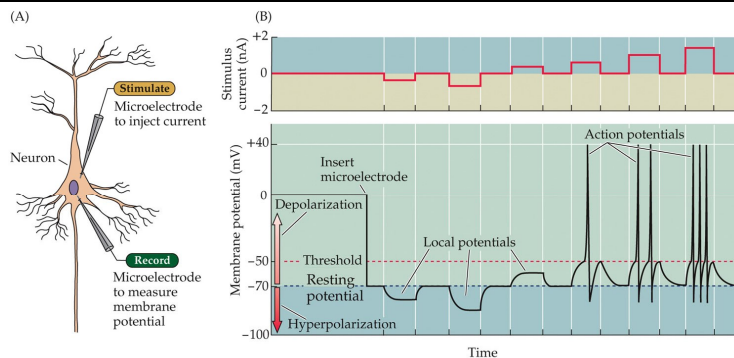
Structure & Function of the Nervous System

Principal parts of neurons



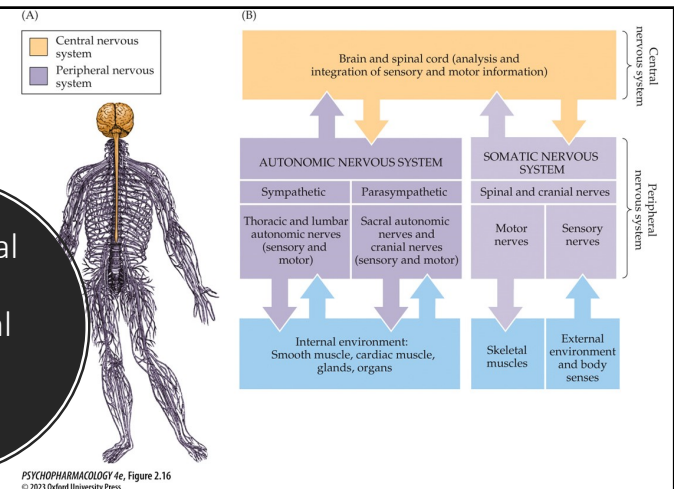
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Local potentials and action potentials



PSYCHOPHARMACOLOGY 4e, Figure 2.12
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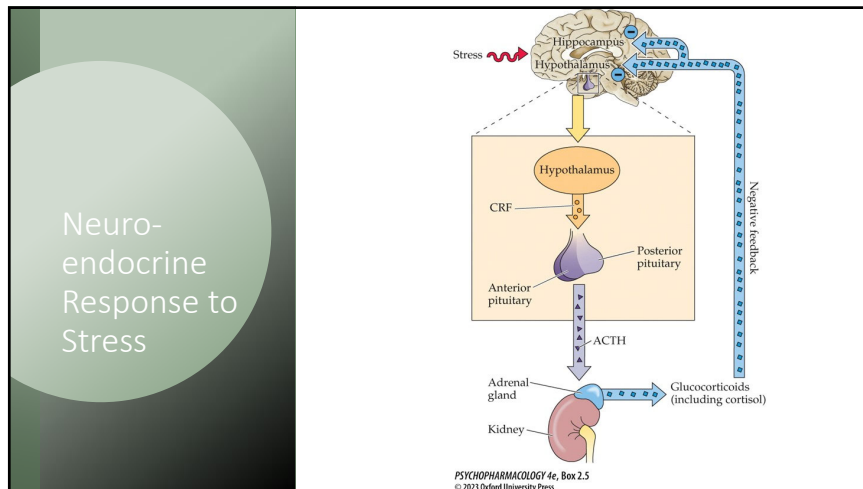
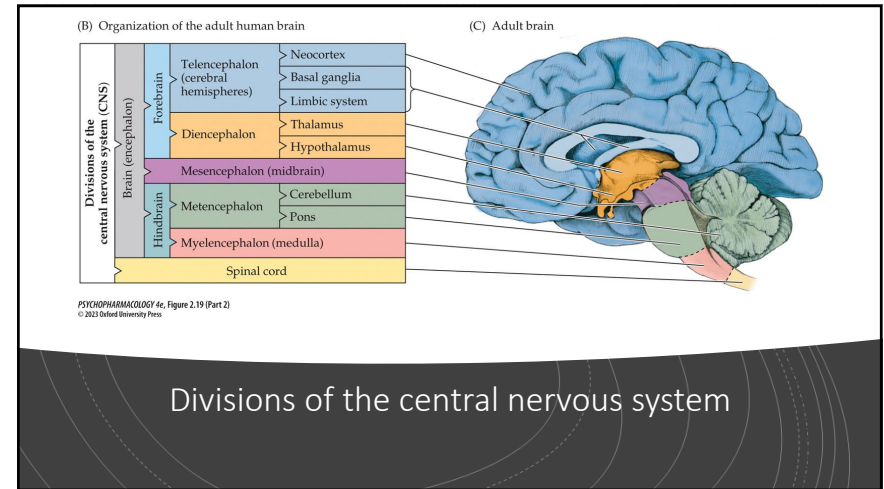
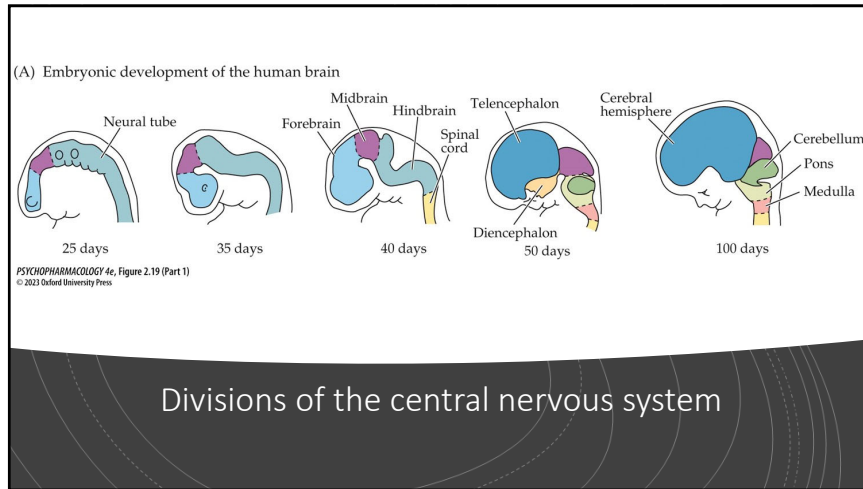
The central and peripheral nervous systems



PSYCHOPHARMACOLOGY 4e, Figure 2.16
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Ch 02 – Structure & Function of the NS

Ch 03 – Chemical Signaling



Chemical Signaling

Learning objectives

General principles of neurotransmission

Synthesis, release and inactivation

Receptors and 2nd messenger systems

Mechanisms by which drugs can alter neurotransmission

All neurons share several structural features

Soma (cell body)

Dendrites & spines

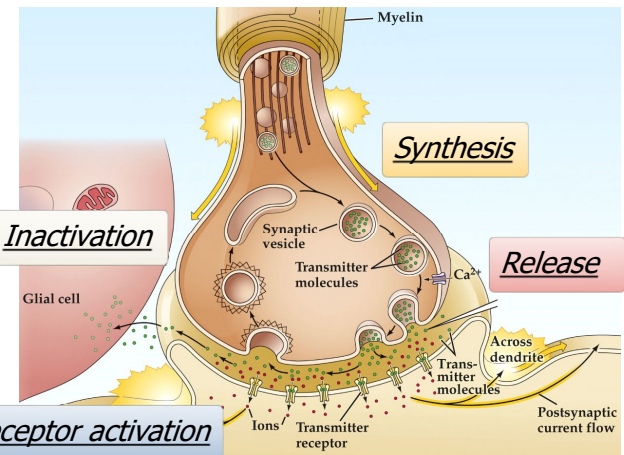
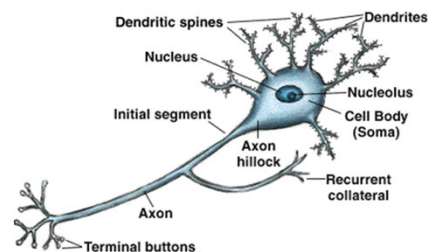
- neurotransmitter receptors

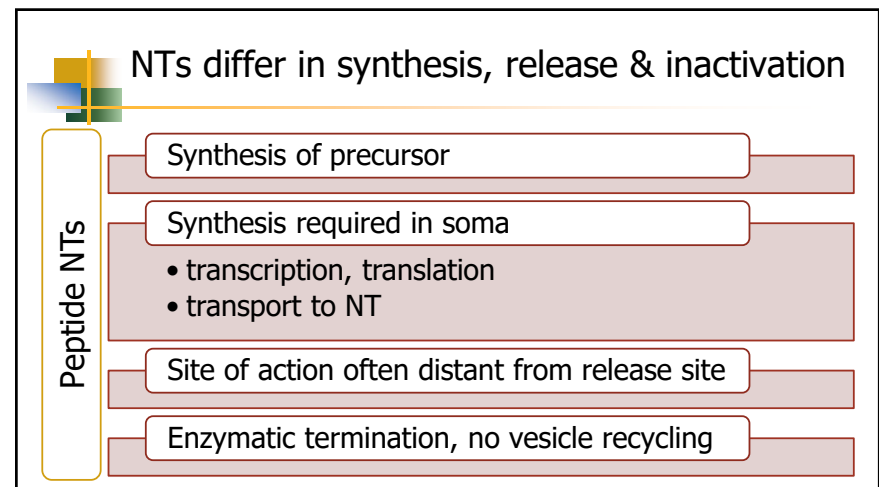
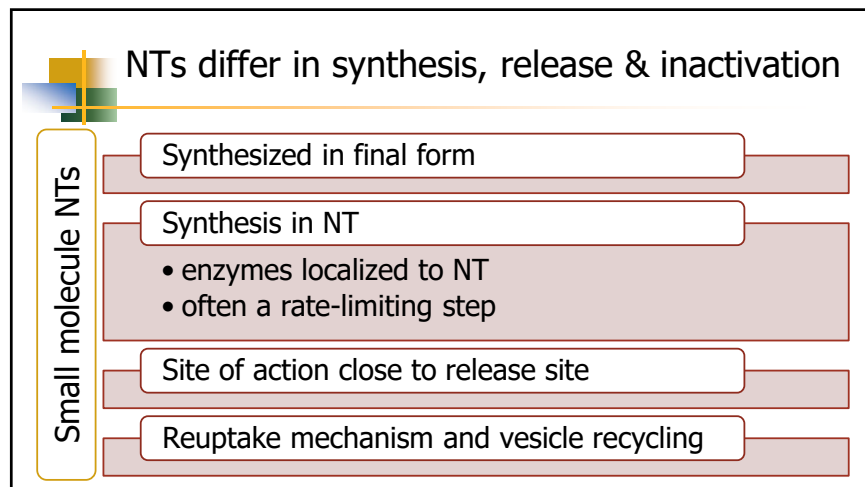
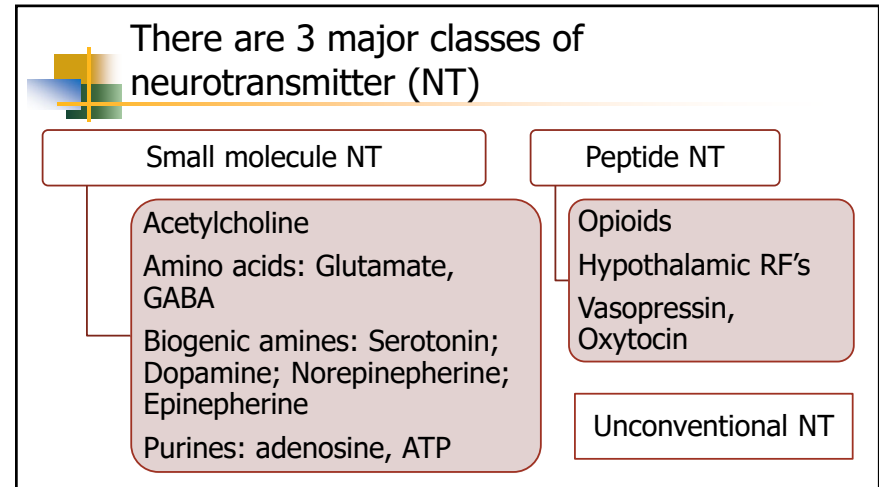
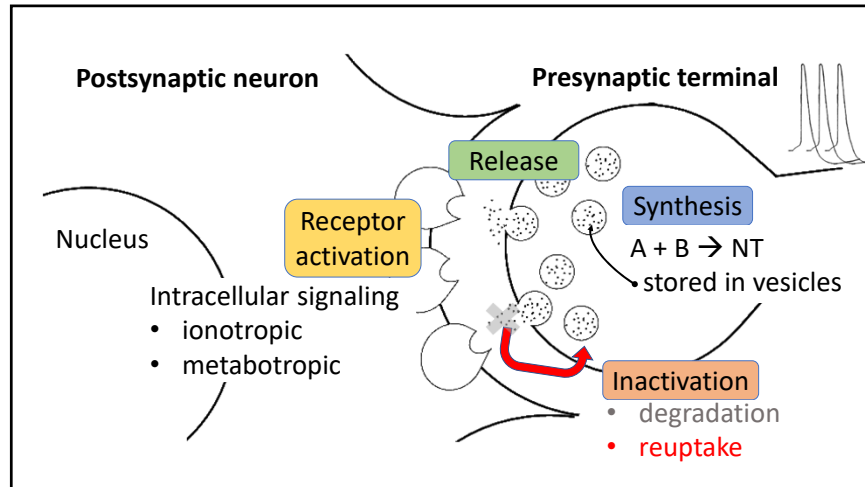
Axon

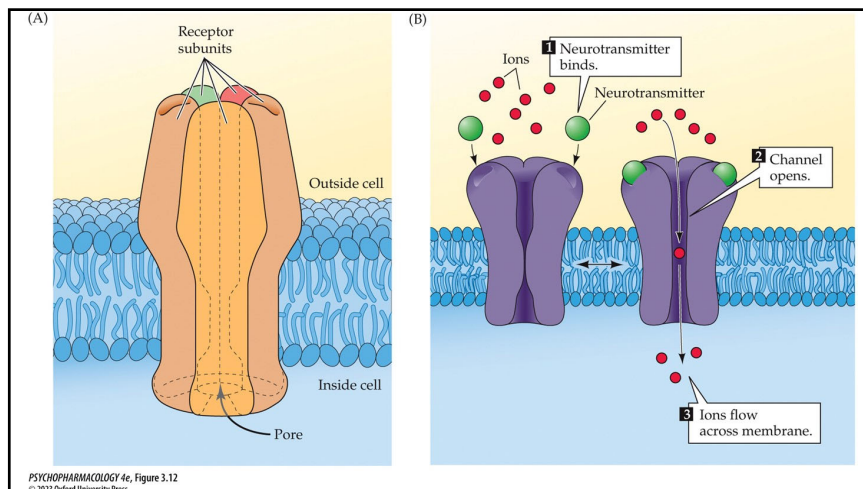
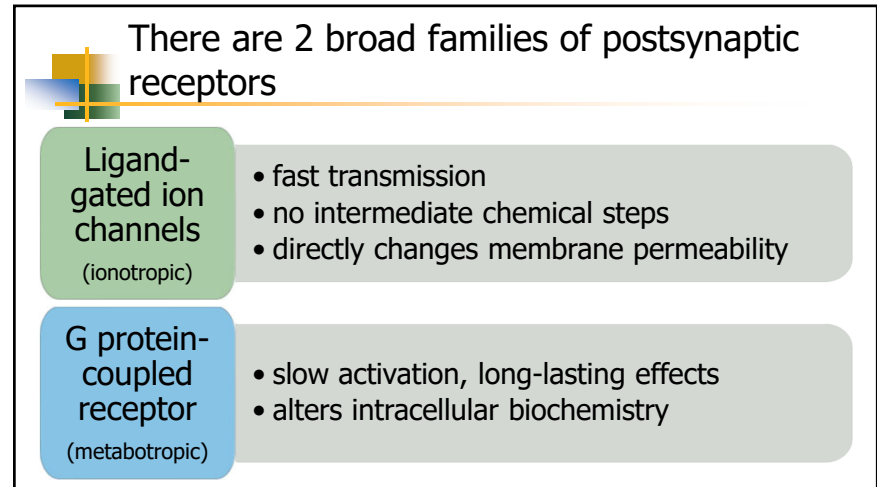
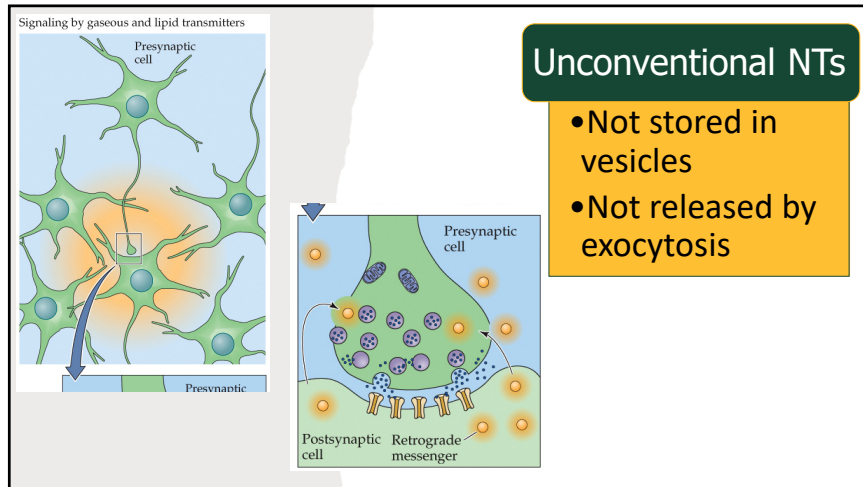
- transmitter release

Synapse

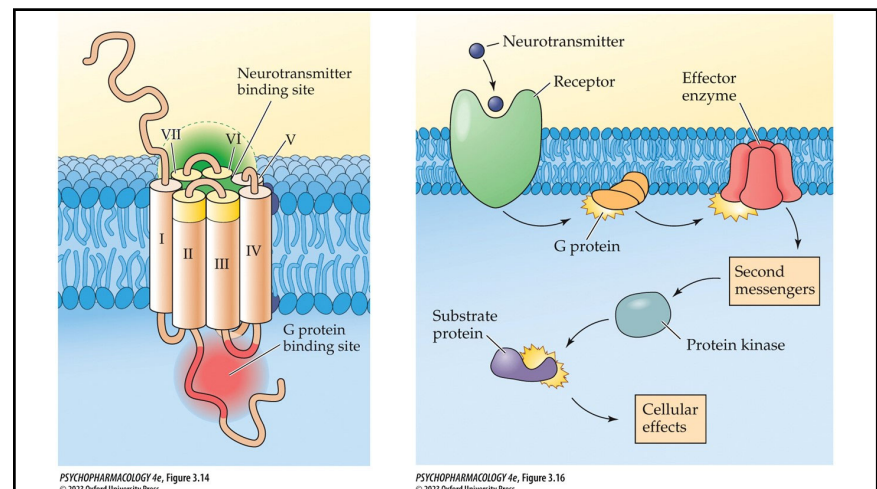
- the gap between neurons







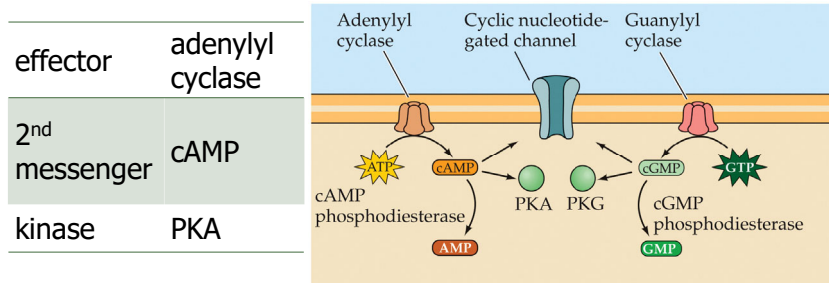
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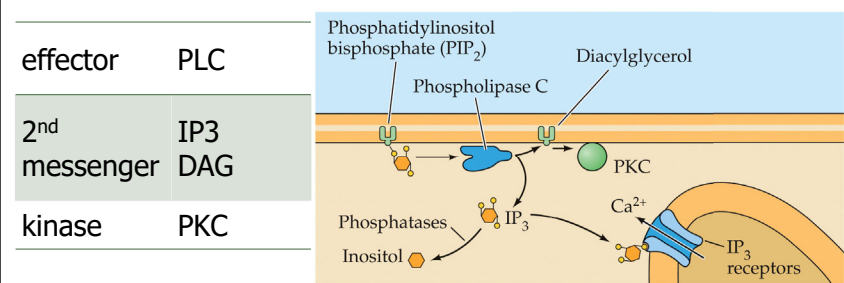
PSYCHOPHARMACOLOGY 4e, Figure 3.14
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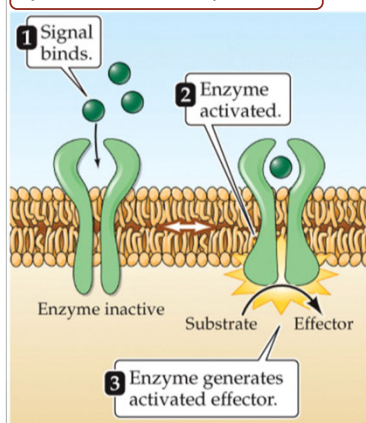
The cyclic AMP system



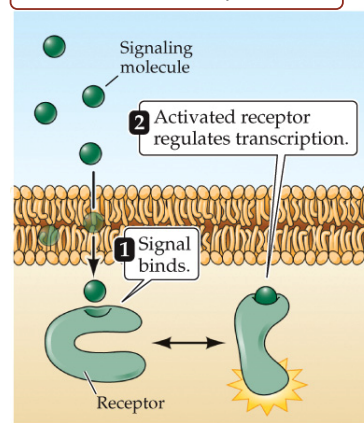
The phospholipase C system



Tyrosine kinase receptors

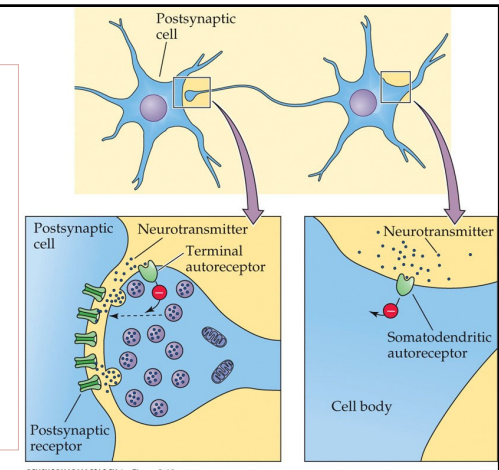


Steroid hormone receptors



Autoreceptors

- A receptor for the same transmitter released by that neuron
- Terminal autoreceptors
 - release-modulating
- Somatodendritic autoreceptors
 - activity modulating



Two primary mechanisms inactivate neurotransmission

Enzymatic breakdown

- In the synapse
AChE
- In the nerve terminal
MAO, COMT
- In blood
peptidases

Reuptake

- Energy dependent pumps move transmitter
 - into the nerve terminal
 - into glia
glutamate, GABA

Psychoactive compounds can affect neurotransmission in several ways

Synthesis

- Increase or decrease
- Neurontin promotes GABA synthesis

Release

AMPH releases DA

Receptors

Thorazine blocks dopamine D2 receptors

Morphine activates opioid receptors

Inactivation

Prozac blocks serotonin reuptake

